

A Crash Course on Bayesian Statistics and its Philosophy

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Contents

1	The Bayesian Recipe	2
2	Examples of Bayesian Inference	3
2.1	Contrast with Maximum Likelihood Estimation	5
2.2	Frequentist and Bayesian Inference Can Disagree	11
2.2.1	Frequentist Analysis	11
2.2.2	Bayesian Analysis	12
2.2.3	The Likelihood Principle	13
3	Coin Fairness Testing and the Likelihood Principle	13
3.1	The Problem	13
3.2	Frequentist Solution: Fixed Sample Size	14
3.3	Frequentist Solution: Stopping After Three Heads	15
3.4	What Changed?	16
3.5	Bayesian Solution	16
4	Interpretations of Probability	17
4.1	The Frequentist Interpretation of Probability	17
4.1.1	Difficulties with the Frequentist Interpretation	18
4.2	The Bayesian Interpretation of Probability	19
4.3	A Coin Tossing Example	19
5	Justification of the Rules of Probability	21
5.1	Plausibilities	22
5.2	The Product Rule	22
5.3	A Consistency Requirement	23
5.4	Solving the Functional Equation	24
5.5	Certainty and Impossibility	24
5.6	The Emergence of Probability	25

6	Frequentist Inference	25
7	Summary	26

1 The Bayesian Recipe

Statistical inference is concerned with learning about unknown quantities from observed data.

In many problems, we model the observed data as being generated from a probability distribution P_θ where $\theta \in \Theta$. Thus, we have a class of probability distributions indexed by a parameter space Θ . The problem is to infer about the underlying θ after observing data. This abstraction is the starting point for simple parametric models all the way up to state of the art ML models like Large Language Models and Diffusion Models.

The Bayesian approach begins with a simple idea.

Before seeing any data, we are uncertain about the value of θ . This uncertainty itself should be represented using probability.

Thus we place a probability distribution on the parameter space Θ .

This distribution is called the *prior distribution* and is denoted by

$$\pi(\theta).$$

The data are modeled through a conditional distribution

$$p(x \mid \theta),$$

called the *likelihood*.

After observing data

$$X = x,$$

Bayes' rule gives the posterior distribution

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\pi(\theta)}{\int_{\Theta} p(x \mid u)\pi(u) du}.$$

The posterior distribution summarizes what we believe about θ after observing the data. The numerator is simply the product of the prior and the likelihood of the observed data, while the denominator is the marginal probability of the observed data, often called evidence in the literature.

$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}.$

Bayesian inference can therefore be summarized in three steps:

1. Specify a prior distribution (Modelling Step)
2. Specify a likelihood (Modelling Step).
3. Apply Bayes' rule.

Everything else in Bayesian statistics follows from these three steps.

2 Examples of Bayesian Inference

Example 2.1 (A Patient Covered in Spots). Suppose a patient arrives at a hospital covered in spots. The symptoms are consistent with either chickenpox or smallpox.

Let

$$\Theta \in \{\text{smallpox}, \text{chickenpox}, \text{neither}\}.$$

Suppose we assign the likelihoods

$$P(\text{spots} \mid \text{smallpox}) = 0.9,$$

$$P(\text{spots} \mid \text{chickenpox}) = 0.8,$$

and

$$P(\text{spots} \mid \text{neither}) = 0.$$

Further suppose that smallpox is extremely rare and we assign the prior

$$P(\text{smallpox}) = 0.001,$$

while

$$P(\text{chickenpox}) = 0.1.$$

This leaves

$$P(\text{neither}) = 0.899.$$

The inference question is:

$$P(\text{smallpox} \mid \text{spots})$$

and

$$P(\text{chickenpox} \mid \text{spots}).$$

To calculate these probabilities, we first compute the overall probability of observing spots.

Using the law of total probability,

$$P(\text{spots}) = P(\text{spots} \mid \text{smallpox})P(\text{smallpox}) + P(\text{spots} \mid \text{chickenpox})P(\text{chickenpox}) + P(\text{spots} \mid \text{neither})P(\text{neither}).$$

Substituting the numerical values gives

$$P(\text{spots}) = 0.9(0.001) + 0.8(0.1) + 0(0.899) = 0.0809.$$

We now apply Bayes' rule.

For smallpox,

$$P(\text{smallpox} \mid \text{spots}) = \frac{P(\text{spots} \mid \text{smallpox})P(\text{smallpox})}{P(\text{spots})}.$$

Substituting the values,

$$P(\text{smallpox} \mid \text{spots}) = \frac{0.9 \times 0.001}{0.0809} \approx 0.011.$$

Similarly,

$$P(\text{chickenpox} \mid \text{spots}) = \frac{P(\text{spots} \mid \text{chickenpox})P(\text{chickenpox})}{P(\text{spots})} = \frac{0.8 \times 0.1}{0.0809} \approx 0.989.$$

Thus

$$P(\text{smallpox} \mid \text{spots}) \approx 0.011,$$

while

$$P(\text{chickenpox} \mid \text{spots}) \approx 0.989.$$

Even though spots are slightly more likely under smallpox than under chickenpox, smallpox is so rare that the posterior probability of chickenpox is overwhelmingly larger.

The Bayesian conclusion is therefore:

The patient almost certainly has chickenpox rather than smallpox.

Remark 2.2. *Notice what happened here.*

The likelihood slightly favors smallpox because

$$P(\text{spots} \mid \text{smallpox}) = 0.9 > 0.8 = P(\text{spots} \mid \text{chickenpox}).$$

However, the prior strongly favors chickenpox because

$$P(\text{chickenpox}) = 0.1 \gg 0.001 = P(\text{smallpox}).$$

Bayes' rule combines these two sources of information.

This example illustrates a central lesson of Bayesian statistics:

The most likely explanation of the data need not be the most plausible explanation after taking prior information into account.

2.1 Contrast with Maximum Likelihood Estimation

A widely used alternative to full Bayesian inference is *maximum likelihood estimation*.

Given observed data x , the maximum likelihood estimator is defined by

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(x \mid \theta).$$

In words, the maximum likelihood estimator chooses the parameter value that makes the observed data most likely.

Returning to the spots example, the observed data are simply that the patient has spots.

The likelihoods are

$$P(\text{spots} \mid \text{smallpox}) = 0.9,$$

and

$$P(\text{spots} \mid \text{chickenpox}) = 0.8.$$

Since

$$0.9 > 0.8,$$

the maximum likelihood estimator chooses

$$\hat{\theta}_{\text{MLE}} = \text{smallpox}.$$

This conclusion is very different from the Bayesian conclusion.

The Bayesian analysis gave

$$P(\text{smallpox} \mid \text{spots}) \approx 0.011,$$

and

$$P(\text{chickenpox} \mid \text{spots}) \approx 0.989.$$

Thus the Bayesian analysis overwhelmingly favors chickenpox.

The reason for this discrepancy is simple.

Maximum likelihood only uses the likelihood:

$$p(x \mid \theta).$$

Bayesian inference uses both the likelihood and the prior:

$$\pi(\theta \mid x) \propto p(x \mid \theta)\pi(\theta).$$

The prior incorporates the fact that smallpox is extraordinarily rare, while maximum likelihood ignores this information entirely.

Conceptually, the two approaches answer different questions.

Maximum likelihood asks:

Which parameter value makes the observed data most likely?

Bayesian inference asks:

Given the observed data, what should we believe about the parameter?

In many problems the answers may be similar. In this example, however, the answers are dramatically different.

This example highlights one of the major strengths of Bayesian inference: it provides a principled way of incorporating prior information into statistical decision making.

Example 2.3 (Six Measurements). Suppose a scientist measures an unknown physical quantity θ six times and obtains

$$26.6, 38.5, 34.4, 34, 31, 23.6.$$

Denote these observations by

$$x_1, \dots, x_n,$$

where

$$n = 6.$$

On the basis of these measurements, what can we infer about θ ?

The Bayesian solution begins with a probabilistic model. We need to specify both a likelihood and a prior distribution.

The likelihood represents the probability of the observed data given parameter values. Besides the unknown quantity θ , we introduce another parameter $\sigma > 0$ which represents the scale of the measurement noise.

Thus our parameter vector is

$$(\theta, \sigma).$$

We shall use the Gaussian likelihood

$$L(\theta, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x_i - \theta)^2}{2\sigma^2} \right\}.$$

At this point it is worth discussing carefully what this likelihood means.

Let

$$X_1, \dots, X_n$$

denote potential measurements. Because measurements are recorded only up to some finite precision, the observed value 26.6 should really be interpreted as belonging to a small interval

$$[26.6 - \delta, 26.6 + \delta]$$

for some small rounding error $\delta > 0$.

Thus the likelihood is formally

$$L(\theta, \sigma) = P\left(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_n \in [x_n - \delta, x_n + \delta] \mid \theta, \sigma\right).$$

For small δ , a probability-density approximation gives

$$L(\theta, \sigma) \approx \delta^n f_{X_1, \dots, X_n \mid \theta, \sigma}(x_1, \dots, x_n).$$

We now assume that

$$f_{X_1, \dots, X_n \mid \theta, \sigma}(x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x_i - \theta)^2}{2\sigma^2} \right\}.$$

The multiplicative factor δ^n is constant with respect to θ and σ , so it can be ignored. This leads to the likelihood above.

A subtle point is worth emphasizing here.

Many texts write the model as

$$X_1, \dots, X_n \mid \theta, \sigma \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2).$$

This is a stronger statement than simply specifying the likelihood at the observed data. Indeed, the i.i.d. normal assumption means that for *every*

$$(u_1, \dots, u_n) \in \mathbb{R}^n,$$

the joint density satisfies

$$f_{X_1, \dots, X_n \mid \theta, \sigma}(u_1, \dots, u_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(u_i - \theta)^2}{2\sigma^2} \right\}.$$

The posterior calculation only uses the likelihood evaluated at the observed data. The full i.i.d. normal assumption specifies the probability law of all possible datasets that could have been observed. The following remark expands on this point.

Remark 2.4 (Likelihood versus full probabilistic model). *It is important to distinguish between the likelihood and a full probabilistic model for the data.*

The assumption

$$X_1, \dots, X_n \mid \theta, \sigma \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2)$$

is equivalent to requiring that

$$f_{X_1, \dots, X_n \mid \theta, \sigma}(u_1, \dots, u_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(u_i - \theta)^2}{2\sigma^2} \right\}$$

for every possible dataset

$$(u_1, \dots, u_n) \in \mathbb{R}^n.$$

This is a very strong assumption.

Indeed, consider the alternative model

$$f_{X_1, \dots, X_n \mid \theta, \sigma}(u_1, \dots, u_n) = \begin{cases} \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(u_i - \theta)^2}{2\sigma^2} \right\}, & (u_1, \dots, u_n) \in (20, 30)^n, \\ \text{arbitrary}, & (u_1, \dots, u_n) \notin (20, 30)^n. \end{cases}$$

This model is clearly not the same as the i.i.d. normal model.

However, if the observed data happen to be

$$(x_1, \dots, x_n) \in (20, 30)^n,$$

then the likelihood function obtained from this alternative model is exactly the same as the Gaussian likelihood.

Consequently, the Bayesian posterior calculation would be identical.

This illustrates that the posterior depends only on the likelihood evaluated at the observed data and not on the full probability model for all possible datasets.

To complete the Bayesian model, we need a prior distribution.

We take

$$\theta \sim \text{Uniform}(-C, C)$$

and

$$\log \sigma \sim \text{Uniform}(-C, C),$$

independently, where C is a very large positive constant.

The idea is that we do not wish to express strong prior information about either parameter.

The corresponding joint prior density is

$$\pi(\theta, \sigma) \propto \frac{\mathbf{1}\{-C < \theta < C, -C < \log \sigma < C\}}{\sigma}.$$

Using Bayes' rule,

$$\pi(\theta, \sigma \mid x_1, \dots, x_n) \propto \pi(\theta, \sigma) L(\theta, \sigma).$$

Substituting the prior and likelihood gives

$$\pi(\theta, \sigma \mid x_1, \dots, x_n) \propto \sigma^{-n-1} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \right\} \mathbf{1}\{-C < \theta < C, -C < \log \sigma < C\}.$$

This is the joint posterior density of θ and σ .

To obtain the posterior density of θ , we integrate out σ :

$$\pi(\theta \mid x_1, \dots, x_n) \propto \mathbf{1}\{-C < \theta < C\} \int_{e^{-C}}^{e^C} \sigma^{-n-1} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \right\} d\sigma.$$

Since C is very large, we may replace the limits of integration by 0 and ∞ .

Define

$$S(\theta) = \sum_{i=1}^n (x_i - \theta)^2.$$

Then

$$\pi(\theta \mid x_1, \dots, x_n) \propto \int_0^\infty \sigma^{-n-1} \exp \left\{ -\frac{S(\theta)}{2\sigma^2} \right\} d\sigma.$$

Using the change of variables

$$t = \frac{S(\theta)}{2\sigma^2},$$

one obtains

$$\pi(\theta \mid x_1, \dots, x_n) \propto [S(\theta)]^{-n/2}.$$

Thus the posterior density is inversely proportional to a power of the residual sum of squares.

Let

$$\hat{\theta} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Since $\hat{\theta}$ minimizes $S(\theta)$, the posterior density is maximized at $\hat{\theta}$.

Using the standard least-squares identity,

$$S(\theta) = S(\hat{\theta}) + n(\theta - \hat{\theta})^2,$$

we obtain

$$\pi(\theta \mid x_1, \dots, x_n) \propto \left(\frac{1}{S(\hat{\theta}) + n(\theta - \hat{\theta})^2} \right)^{n/2}.$$

Equivalently,

$$\pi(\theta \mid x_1, \dots, x_n) \propto \left(1 + \frac{n(\theta - \hat{\theta})^2}{S(\hat{\theta})} \right)^{-n/2}.$$

This is exactly a Student- t density. In fact,

$$\frac{\sqrt{n}(\theta - \hat{\theta})}{\sqrt{S(\hat{\theta})/(n-1)}} \bigg| x_1, \dots, x_n \sim t_{n-1}.$$

Thus the posterior distribution is centered at the sample mean.

The posterior mean, median and mode are all equal to

$$\hat{\theta} = \bar{x}.$$

A $100(1 - \alpha)\%$ Bayesian credible interval for θ is therefore

$$\left[\hat{\theta} - \frac{t_{n-1, \alpha/2}}{\sqrt{n}} \sqrt{\frac{S(\hat{\theta})}{n-1}}, \hat{\theta} + \frac{t_{n-1, \alpha/2}}{\sqrt{n}} \sqrt{\frac{S(\hat{\theta})}{n-1}} \right].$$

For $\alpha = 0.05$, this gives the interval

$$[25.598, 37.102].$$

Remark 2.5 (Bayesian versus Frequentist Interpretation). *An interesting feature of this example is that the Bayesian credible interval is numerically identical to the classical Student-t confidence interval. However, the interpretations are different.*

The frequentist interpretation is:

If we repeated the experiment many times and constructed a confidence interval each time, then approximately 95% of those intervals would contain the true value of θ .

The Bayesian interpretation is:

Given the observed data, the posterior probability that θ lies in the interval is 0.95.

2.2 Frequentist and Bayesian Inference Can Disagree

In the previous example, the Bayesian credible interval and the frequentist confidence interval happened to coincide. The agreement in the previous example is nice but coincidental and it is easy to break this coincidence. A slight change in the experimental design can lead to different conclusions.

Consider the following modification of the measurement problem.

Example 2.6 (Measurements with a Stopping Rule). A scientist repeatedly measures an unknown physical quantity θ until a measurement smaller than 25 is obtained.

Suppose the resulting measurements are

$$26.6, ; 38.5, ; 34.4, ; 34, ; 31, ; 23.6.$$

Should our inference about θ change?

Notice that the observed data are exactly the same as in the previous example. The only difference is the manner in which the data were collected.

2.2.1 Frequentist Analysis

The confidence interval from the previous example is no longer automatically valid. The reason is that the sample size is no longer fixed. Instead, the number of observations becomes the random variable

$$N = \inf\{n \geq 1 : X_n \leq 25\}. \quad (1)$$

The confidence interval from the previous example was based on the probability statement

$$\frac{\sqrt{n}(\bar{X}_n - \theta)}{\hat{\sigma}} \sim t_{n-1},$$

which assumes that the sample size is deterministically equal to n .

In the present problem, the relevant probability becomes

$$P \left(\bar{X}_N - \frac{t_{N-1, \alpha/2}}{\sqrt{N}} \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2} \leq \theta \leq \bar{X}_N + \frac{t_{N-1, \alpha/2}}{\sqrt{N}} \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2} \right).$$

Here the probability is computed under the model

$$X_1, X_2, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2),$$

where N is the random variable in (1).

There is no reason for the probability above to be exactly equal to $1 - \alpha$. In other words, the stopping rule changes the repeated sampling distribution of the procedure.

Constructing valid frequentist confidence intervals in the presence of stopping rules is a subtle problem and remains an active area of research.

2.2.2 Bayesian Analysis

In contrast to frequentist inference, the Bayesian inference procedure will not change. This is because the likelihood function in the present problem is exactly the same as the likelihood function in the original fixed-sample-size problem.

To verify this, let δ denote the (very small) rounding error in the measurements. The likelihood in the present problem is

$$\begin{aligned} \text{Likelihood} &= P(\text{observed data} \mid \theta, \sigma) \\ &= P(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_6 \in [x_6 - \delta, x_6 + \delta], \quad X_1 \geq 25, \dots, X_5 \geq 25, X_6 < 25 \mid \theta, \sigma). \end{aligned}$$

In other words, we are writing the probability that the first five observations are all larger than 25 while the sixth observation is smaller than 25, in addition to the exact observed values.

However, these additional constraints do not change the probability.

As a simple example, note that $P(Z = 5) = P(Z = 5, Z < 10)$. The additional restriction $Z < 10$ does not affect the probability because it is already implied by the event $Z = 5$.

Similarly,

$$\begin{aligned} \text{Likelihood} &= P\left(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_6 \in [x_6 - \delta, x_6 + \delta], \right. \\ &\left. X_1 \geq 25, \dots, X_5 \geq 25, X_6 < 25 \mid \theta, \sigma\right) = P\left(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_6 \in [x_6 - \delta, x_6 + \delta] \mid \theta, \sigma\right). \end{aligned}$$

But the expression on the right-hand side is exactly the likelihood from the original problem. Therefore,

$$\text{Likelihood in Problem 1} = \text{Likelihood in Problem 2}.$$

Since the likelihood is the same in both problems, and since there is no reason to change the prior distribution, the posterior distribution is also exactly the same.

Consequently, all Bayesian conclusions remain unchanged. The posterior mean, posterior mode, posterior credible interval, and indeed the entire posterior distribution are identical to those obtained in the original problem.

Therefore, from the Bayesian perspective, stopping rules can be ignored for inference about θ , because they do not affect the likelihood.

2.2.3 The Likelihood Principle

The likelihood principle states that all evidence in the data relevant to the unknown parameters is contained in the likelihood function. The two problems considered above have exactly the same likelihood function. A Bayesian therefore reaches exactly the same conclusion in both cases. A frequentist, on the other hand, generally reaches different conclusions because frequentist procedures depend not only on the observed data but also on the sampling plan and stopping rule. Thus standard frequentist inference violates the likelihood principle, whereas Bayesian inference automatically satisfies it.

3 Coin Fairness Testing and the Likelihood Principle

We now consider another example where frequentist and Bayesian reasoning can lead to different conclusions. The example concerns testing whether a coin is fair.

3.1 The Problem

Suppose a coin is tossed 12 times and the observed sequence is

$$TTTTHTHTTTTH.$$

Thus there are 3 heads and 9 tails.

The question is:

Is there evidence that the coin is biased against heads?

Let p denote the probability that the coin lands heads. The null hypothesis of fairness is

$$H_0 : p = \frac{1}{2}.$$

A natural one-sided alternative is

$$H_1 : p < \frac{1}{2}.$$

We now compare two frequentist analyses of exactly the same observed data.

3.2 Frequentist Solution: Fixed Sample Size

First suppose that the experiment was designed in advance as follows:

Toss the coin exactly 12 times.

Let X denote the number of heads in 12 tosses. Under the null hypothesis $H_0 : p = 1/2$, we have

$$X \sim \text{Binomial}\left(12, \frac{1}{2}\right).$$

The observed value is

$$X_{\text{obs}} = 3.$$

To test against the alternative $p < 1/2$, the p-value is the probability, under the null hypothesis, of observing 3 or fewer heads:

$$\text{p-value} = P_{p=1/2}(X \leq 3).$$

Therefore

$$\begin{aligned} P_{p=1/2}(X \leq 3) &= \sum_{k=0}^3 \binom{12}{k} \left(\frac{1}{2}\right)^{12} \\ &= \left[\binom{12}{0} + \binom{12}{1} + \binom{12}{2} + \binom{12}{3} \right] \left(\frac{1}{2}\right)^{12} \\ &= \frac{1 + 12 + 66 + 220}{4096} \\ &= \frac{299}{4096} \\ &\approx 0.073. \end{aligned}$$

Thus the p-value is approximately 7.3%. At the usual 5% significance level, we would not reject the null hypothesis that the coin is fair.

3.3 Frequentist Solution: Stopping After Three Heads

Now suppose instead that the experiment was designed differently:

Toss the coin until three heads have appeared.

The observed sequence is still

$$TTTTHTHTTTTH,$$

so the third head appeared on the 12th toss.

Let N denote the number of tosses needed to obtain three heads. Under the null hypothesis $p = 1/2$, N has a negative binomial distribution.

The observed value is

$$N_{\text{obs}} = 12.$$

Evidence against fairness in the direction $p < 1/2$ corresponds to needing an unusually large number of tosses to obtain three heads. Therefore the p-value is

$$P_{p=1/2}(N \geq 12).$$

Equivalently,

$$P_{p=1/2}(N \geq 12) = 1 - P_{p=1/2}(N \leq 11).$$

For $n \geq 3$, the probability that the third head occurs exactly on toss n is

$$P_{p=1/2}(N = n) = \binom{n-1}{2} \left(\frac{1}{2}\right)^n.$$

Indeed, among the first $n - 1$ tosses there must be exactly two heads, and the n th toss must be a head.

Thus

$$\begin{aligned} P_{p=1/2}(N \geq 12) &= 1 - \sum_{n=3}^{11} \binom{n-1}{2} \left(\frac{1}{2}\right)^n \\ &= \frac{134}{4096} \\ &\approx 0.0327. \end{aligned}$$

So in this second analysis the p-value is approximately 3.27%. At the usual 5% significance level, we would reject the null hypothesis that the coin is fair.

3.4 What Changed?

The observed sequence of tosses is exactly the same in the two analyses. In both cases we observed

$$TTTTHTHTTTTH,$$

which contains 3 heads and 9 tails.

However, the frequentist p-value changed from

$$0.073$$

to

$$0.0327.$$

The reason is that frequentist p-values are computed using the sampling distribution of the test statistic under the null hypothesis. This sampling distribution depends not only on the observed data, but also on the experimental design.

If the number of tosses was fixed in advance, the relevant null distribution is binomial. If the experiment was stopped after three heads appeared, the relevant null distribution is negative binomial.

Thus, under frequentist testing, the conclusion depends on the stopping rule.

Remark 3.1. *The sample space changed, in the first case, the sample space is the set of sequences of H and T of length 12. In the second case, the sample space is the set of sequences of arbitrary length finishing with the third H. Probability assignment in the sample space has also naturally changed. When we calculate p values, we are calculating probabilities of the observed sequence plus some other hypothetical sequences that we have not observed. This is the reason the p value changed.*

3.5 Bayesian Solution

From the Bayesian perspective, the inference does not depend on whether the experimenter decided in advance to toss the coin 12 times or instead decided to continue tossing until three heads were observed.

The reason is that the likelihood function is identical in both situations.

In the first experiment, the observed data consist of three heads and nine tails. Ignoring multiplicative constants that do not depend on p , the likelihood is

$$L(p) \propto p^3(1-p)^9.$$

In the second experiment, the coin is tossed until the third head appears. The probability of observing the particular sequence

TTTTHTHTTTTH

is again proportional to

$$L(p) \propto p^3(1-p)^9.$$

Thus the two experiments lead to exactly the same likelihood function.

If the prior distribution for p is also taken to be the same in both experiments, then Bayes' rule gives

$$\pi(p \mid \text{data}) \propto L(p)\pi(p),$$

and therefore the posterior distribution is identical in the two cases.

Consequently, all Bayesian conclusions are unchanged: the posterior mean, posterior mode, credible intervals, posterior probabilities, and any Bayesian decision rule will be exactly the same.

This is another illustration of the likelihood principle. The two experiments produce the same likelihood and therefore a Bayesian reaches the same conclusion in both cases. The frequentist p-value, however, changes from 7.3% to 3.27% because the sampling plan changes. Thus standard p-value based inference depends on aspects of the experiment beyond the observed likelihood, whereas Bayesian inference depends only on the likelihood and the prior.

4 Interpretations of Probability

Because Bayesian statistics is simply probability theory applied to inference, understanding the meaning and interpretation of probability is essential. There are broadly two ways of understanding probability:

1. The *frequentist* or *objective* interpretation.
2. The *Bayesian* or *subjective* interpretation.

The distinction between these viewpoints is one of the fundamental philosophical differences between Bayesian and frequentist statistics.

4.1 The Frequentist Interpretation of Probability

From the frequentist viewpoint, probability is applicable only in the context of random experiments such as tossing coins, rolling dice, or repeatedly measuring physical quantities.

The probability of an event A is defined as the limiting relative frequency with which A occurs in repeated trials of an experiment:

$$P(A) = \lim_{N \rightarrow \infty} \frac{N_A}{N},$$

where N_A denotes the number of times that event A occurs among N repetitions of the experiment.

According to this interpretation, probability is fundamentally a statement about long-run frequencies.

Suppose that H denotes the event that a coin lands heads.

The statement

$$P(H) = 0.5$$

means that the proportion of heads in a large number of tosses approaches 0.5.

As another example, consider the statement

$$\varepsilon_1, \dots, \varepsilon_n \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

This means that if the experiment generating $(\varepsilon_1, \dots, \varepsilon_n)$ is repeated many times, then for every set $A \subseteq \mathbb{R}^n$,

$$\frac{\text{replications with } (\varepsilon_1, \dots, \varepsilon_n) \in A}{\text{number of replications}} \rightarrow \int_A \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x_i^2}{2\sigma^2}\right) dx_1 \cdots dx_n.$$

Thus probability is interpreted as a property of repeated experiments.

4.1.1 Difficulties with the Frequentist Interpretation

Although intuitively appealing, this interpretation faces a number of difficulties.

Problem 1: Many probability statements do not involve repeated experiments. For example, consider the questions:

1. Is the suspect guilty?
2. What is the chance of rain in Bengaluru tomorrow?
3. What is the probability that a patient has cancer given a positive test result?

In these situations, it is not obvious what repeated experiment is being performed.

The frequency interpretation therefore has difficulty assigning a meaning to such probability statements.

Problem 2: What does it mean to repeat an experiment identically? Consider a coin.

The statement

$$P(H) = 0.6$$

is supposed to mean that approximately 60 percent of a large number of tosses result in heads. But no two tosses are ever exactly identical. Indeed, if two tosses were performed under precisely the same physical conditions, then classical mechanics would suggest that they should produce the same outcome.

Thus the phrase

“independent and identical repetitions of an experiment”

is itself somewhat ambiguous.

4.2 The Bayesian Interpretation of Probability

The Bayesian viewpoint takes a fundamentally different approach to probability.

Rather than viewing probability as a statement about long-run frequencies in repeated experiments, Bayesian statistics treats probability as a general language for reasoning under uncertainty. Probability is therefore applicable whenever uncertainty is present and is not restricted to situations involving random experiments.

In particular, Bayesian probability need not be connected to frequency. Different analysts may use different probability models, reflecting different states of information or different modeling assumptions. These models can then be judged according to their usefulness, predictive performance, or within a Bayesian model comparison framework.

A probability statement can therefore be interpreted without reference to any hypothetical sequence of repeated experiments.

For example, suppose a physician assigns probability

$$P(\text{Cancer}) = 0.02$$

to the event that a patient has cancer. Following the discussion in Lindley’s book *Understanding Uncertainty* [Lindley2013], this statement may be interpreted as follows.

The physician’s uncertainty about whether the patient has cancer is equivalent to the uncertainty associated with drawing a single ball from an urn containing two red balls and ninety-eight green balls.

The crucial point is that no notion of repeated sampling is involved. The urn is sampled once and only once. The long-run frequency of red balls in repeated drawings is irrelevant. The probability statement concerns uncertainty about a single event.

4.3 A Coin Tossing Example

Consider the probability assignment

$$P(H \mid I) = 0.5,$$

where H denotes the event that the next toss of a coin lands heads and I denotes some background information.

In the Bayesian framework, this statement is not a claim about the limiting frequency of heads in an infinite sequence of tosses. Rather, it expresses a state of belief regarding the outcome of the next toss, given the information I .

The same probability assignment can arise from very different kinds of information.

For example, suppose that I_1 represents the following information:

We know essentially nothing about the coin. We do not even know whether the coin has one head side and one tail side. Furthermore, we do not know how the coin will be tossed.

Now suppose that I_2 represents a completely different situation:

We know that the coin is an ordinary coin with one head side and one tail side, and that it will be tossed in the usual manner.

Both information states may reasonably lead to the assignment

$$P(H \mid I_1) = P(H \mid I_2) = 0.5.$$

Notice, however, that in the first situation it is entirely possible that the coin actually has two head sides. If this were the case, repeated tossing would produce a sequence

$$H, H, H, H, \dots$$

and the long-run frequency of heads would be one rather than one-half.

This does not invalidate the probability assignment

$$P(H \mid I_1) = 0.5,$$

because the assignment was never intended to be a statement about long-run frequencies. It was a statement about uncertainty concerning the next toss.

Suppose now that we have additional information.

Assume that the coin has already been tossed many times and that approximately 70 percent of the observed outcomes were heads.

The quantity of interest then becomes

$$P\left(X_{n+1} = 1 \mid \frac{X_1 + \dots + X_n}{n} = 0.7\right),$$

where $X_i = 1$ denotes a head and $X_i = 0$ denotes a tail.

In this setting, the probability assignment can no longer be made arbitrarily. Instead, it must be derived from a probabilistic model describing the joint behavior of

$$X_1, \dots, X_n, X_{n+1}.$$

To illustrate this point, consider two different models.

The first model assumes

$$X_1, \dots, X_{n+1} \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5).$$

Under this model, X_{n+1} is independent of the previous observations. Consequently,

$$P(X_{n+1} = 1) = 0.5,$$

regardless of the observed frequency of heads in the past.

Now consider a second model.

Assume

$$X_1, \dots, X_{n+1} \mid \theta \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta),$$

and place the prior

$$\theta \sim \text{Uniform}(0, 1).$$

In this model, the previous observations provide information about the unknown parameter θ . When n is large and approximately 70 percent of the observed tosses are heads, the predictive probability of a head on the next toss will be close to 0.7.

Thus Bayesian probability remains distinct from frequency, but an appropriate probabilistic model can naturally lead to conclusions that reflect observed frequencies.

This flexibility is one of the strengths of the Bayesian framework.

5 Justification of the Rules of Probability

In the previous section, we argued that probability can be viewed as a general language for reasoning under uncertainty and need not be interpreted in terms of long-run frequencies. This raises an important question.

If probability is not defined through frequencies, why should it satisfy the familiar rules of probability? Why should probabilities obey the product rule and the sum rule?

This question was raised by many authors. In particular, R. A. Fisher argued that it is difficult to justify the laws of probability without ultimately relying on frequency interpretations. In a famous paper [Fisher1934], he criticized attempts to derive the rules of probability purely from logical principles.

A remarkable answer was given by the physicist R. T. Cox. Cox showed that the ordinary rules of probability can be derived from simple requirements of logical consistency and do not require any appeal to long-run frequencies. These ideas were later developed extensively by Jaynes in his book *Probability Theory: The Logic of Science* [Jaynes2003].

The argument that follows is due to Cox [Cox2001]. We shall follow the presentation in Chapters 1 and 2 of Jaynes [Jaynes2003]. The treatment here is only a sketch; several technical details are omitted.

5.1 Plausibilities

Let us temporarily forget about probabilities and instead work with more general quantities which we shall call *plausibilities*.

The plausibility of proposition A given proposition B will be denoted by

$$(A|B).$$

At this stage we impose very few restrictions. We do not require plausibilities to lie between 0 and 1. We merely assume that they take values in the real numbers and that larger values correspond to stronger belief.

The question is:

What properties must plausibilities satisfy if they are to provide a logically consistent method for reasoning under uncertainty?

5.2 The Product Rule

Consider three propositions A , B , and C .

We would like to determine the plausibility that both A and B are true given C :

$$(AB|C).$$

Here AB denotes the proposition that both A and B are true.

A natural way to decide whether AB is true is to proceed in two stages.

First, assess the plausibility of B given C :

$$(B|C).$$

Then, having accepted B as true, assess the plausibility of A given both B and C :

$$(A|BC).$$

It is therefore reasonable to assume that $(AB|C)$ is completely determined by these two quantities.

Accordingly, we assume the existence of a function F such that

$$(AB|C) = F((B|C), (A|BC)).$$

The same function F should be used for all propositions.

In particular, we may also write

$$(AB|C) = F((A|C), (B|AC)).$$

We further assume that F is continuous and monotone increasing in each argument.

Continuity is natural because a small change in one plausibility should not lead to a large jump in another plausibility. Monotone increasing is also a natural assumption here.

5.3 A Consistency Requirement

Now consider four propositions A , B , C , and D . The plausibility of ABC given D can be computed in two different ways.

We first combine B and C . This gives

$$(ABC|D) = F((BC|D), (A|BCD)).$$

Applying the definition of F to $(BC|D)$ yields

$$(ABC|D) = F\left(F((C|D), (B|CD)), (A|BCD)\right).$$

Alternatively, we may first combine A and B .

This gives

$$(ABC|D) = F((C|D), (AB|CD)).$$

Applying the definition of F again yields

$$(ABC|D) = F\left((C|D), F((B|CD), (A|BCD))\right).$$

Both calculations refer to exactly the same plausibility.

A basic requirement of logical consistency therefore demands that they produce the same answer.

Hence,

$$F\left(F((C|D), (B|CD)), (A|BCD)\right) = F\left((C|D), F((B|CD), (A|BCD))\right).$$

If the individual plausibilities can vary freely, this implies that the function F must satisfy

$$F(F(x, y), z) = F(x, F(y, z))$$

for all real numbers x , y , and z .

5.4 Solving the Functional Equation

It now turns out the only coordinatewise increasing and continuous functions F which satisfy the above equation are of the form

$$F(x, y) = w^{-1}(w(x)w(y)),$$

where w is a positive continuous increasing function.

We shall not prove this result here. A derivation can be found in Chapter 2 of Jaynes [Jaynes2003].

Substituting this representation into our expression for $(AB|C)$ gives

$$(AB|C) = w^{-1}(w(B|C)w(A|BC)).$$

Applying w to both sides yields

$$w(AB|C) = w(B|C)w(A|BC).$$

5.5 Certainty and Impossibility

We now investigate some consequences of this formula.

Set $B = A$. Then

$$w(A|C) = w(A|C)w(A|AC).$$

The proposition A is certainly true once we know that A is true.

Thus $(A|AC)$ corresponds to certainty. The previous equation therefore implies

$$w(\text{certainty}) = 1.$$

Next set $B = A^c$.

Then

$$w(AA^c|C) = w(A^c|C)w(A|A^cC).$$

Both AA^c and $A|A^cC$ correspond to impossibility.

Consequently,

$$w(\text{impossible}) = 0.$$

Since w is increasing, it follows that

$$0 \leq w(A|B) \leq 1$$

for every pair of propositions A and B .

5.6 The Emergence of Probability

Define

$$P(A|B) = w(A|B).$$

The transformed plausibilities now take values between 0 and 1.

Moreover, they satisfy

$$P(AB|C) = P(B|C)P(A|BC).$$

By symmetry,

$$P(AB|C) = P(A|C)P(B|AC).$$

These are precisely the usual product rules of probability.

Remark 5.1. *Once we accept that plausibility of AB = some function F applied to plausibility of A and plausibility of $B|A$ then the proof starts. The proof basically looks at ABC and breaks it into two ways. First BC and then A/BC and then break BC as C and $B|C$. Or first, C and $AB|C$ then break $AB|C$ as $B|C$ and $A|BC$. The plausibilities need to be the same for both these ways. Amazingly, this observation restricts the plausibility function to be essentially a product form up to scaling by a positive increasing function denoted by w in the proof. Note that whatever the F is, there exists a w . This w seems to be special; once we apply w to the plausibilities then we get the usual rules of probability.*

We have therefore shown that the product rule does not arise from repeated experiments or long-run frequencies. Instead, it follows from the requirement that reasoning under uncertainty be logically consistent. The next steps are to derive the sum rule and the complement rule. We leave this as exercises. This will complete Cox's justification of the standard axioms of probability.

6 Frequentist Inference

Frequentist inference takes a different perspective. The parameter θ is viewed as a fixed but unknown quantity, representing the true state of nature. The goal is to construct a statistic $\hat{\theta}$ from the observed data. This estimator may arise from maximum likelihood, penalized likelihood, moment methods, or other principled—and sometimes even ad hoc—considerations.

Once an estimator has been proposed, the central object of study becomes its *sampling distribution*: the distribution of $\hat{\theta}$ when the entire experiment is repeated many times under the same parameter value θ . One then investigates properties such as bias, variance, confidence intervals, and

hypothesis tests. A good estimator is one whose sampling distribution is sufficiently concentrated around the true value θ , uniformly over all possible truths $\theta \in \Theta$.

Much of modern frequentist theory is asymptotic in nature. One considers a sequence of experiments indexed by the sample size n and studies the corresponding sequence of sampling distributions. The resulting analysis seeks to establish properties such as consistency, rates of convergence, asymptotic normality, and minimax optimality.

These guarantees are statements about the long-run behavior of a statistical procedure when the experiment is repeated many times. They do not directly provide a probability statement about the particular value of θ in the single dataset observed by a statistician. From the frequentist perspective, probability is attached to the randomness of the data and of the estimator, not to the unknown parameter itself.

7 Summary

Bayesian statistics provides a remarkably simple and unified framework for inference. A wide variety of statistical and machine learning problems can be treated within a common framework. For example, the same Bayesian machinery can be used for parameter estimation, hypothesis testing, model selection, prediction, regression, classification, reinforcement learning, and probabilistic generative modeling. From this perspective, Bayesian statistics is conceptually very attractive. One does not need to invent different inferential procedures for different problems. Instead, one specifies a probabilistic model, assigns a prior distribution, and computes the posterior distribution. All subsequent inference follows from the posterior.

At the same time, Bayesian methods face important practical challenges. First, Bayesian inference requires specification of a prior distribution. In simple problems, prior information can be extremely useful. However, in high-dimensional problems, choosing an appropriate prior can be difficult. A poorly chosen prior may lead to poor inference or undesirable behavior. Although the influence of the prior often diminishes as the amount of data grows, this phenomenon can be much less pronounced in modern high-dimensional settings.

Second, Bayesian computation is often challenging. The posterior distribution is usually given by

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\pi(\theta)}{\int_{\Theta} p(x \mid u)\pi(u) du},$$

and this quantity can rarely be calculated explicitly.

Only in a handful of special situations, such as certain Gaussian models and conjugate-prior models, can the posterior distribution be obtained in closed form. In most modern applications, one must rely on computational methods such as Markov Chain Monte Carlo (MCMC), Gibbs sampling, Hamiltonian Monte Carlo, Variational Inference, and Sequential Monte Carlo. These methods can be computationally expensive, especially when the parameter dimension is large.

In contrast, many frequentist procedures are computationally much simpler. For example, maximum likelihood estimation only seeks the optimizer

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(x \mid \theta),$$

rather than the entire posterior distribution. Optimization can be considerably easier than integration or sampling, and this computational advantage is one reason why frequentist methods such as maximum likelihood is employed in cutting edge ML models.

To summarize, both Bayesian and frequentist ideas play central roles in statistics and machine learning. The Bayesian tradition emphasizes coherent probabilistic reasoning and uncertainty quantification. The frequentist tradition emphasizes long-run guarantees, robustness, and computational tractability. Neither viewpoint completely dominates the other. Indeed, many modern methods combine ideas from both traditions.

For this reason, it is important to understand not only the mathematical techniques associated with each school of thought, but also the philosophical principles that motivate them.